

Mark Rober's CrunchLabs Beating Carnival Scams With Science

Time	Subtitle	Human Translation
4s	Eleven feet.	Eleven feet.
7s	Eleven feet.	Eleven feet.
9s	It's no secret that carnivals exist to make money.	It's no secret that carnivals exist to make money.
11s	To do that most effectively, they employ a bunch of little tricks	To do that most effectively, they employ a bunch of little tricks
15s	to make you overestimate your chances of winning,	to make you overestimate your chances of winning,
18s	in some cases to such an extent that it's basically a scam.	in some cases to such an extent that it's basically a scam.
21s	I spent a couple of days coming to this amusement park,	I spent a couple of days coming to this amusement park,
24s	observing and collecting data on all of the games.	observing and collecting data on all of the games.
27s	I'm gonna tell you which games are the biggest rip-offs	I'm gonna tell you which games are the biggest rip-offs
29s	and the scientific reasons why.	and the scientific reasons why.
32s	I even Nancy Drew-ed how much the carnival pays	I even Nancy Drew-ed how much the carnival pays
34s	for the prizes people can win.	for the prizes people can win.
36s	But there is hope.	But there is hope.
37s	I'll show you some legitimate tricks on how to win the most popular games,	I'll show you some legitimate tricks on how to win the most popular games,
41s	along with what happens when you show up	along with what happens when you show up
43s	with your baseball-playing buddy who happens to play for the New York Mets.	with your baseball-playing buddy who happens to play for the New York Mets.
52s	Let's get started.	Let's get started.
1:05	Any legitimate investigation	Any legitimate investigation
1:07	needs to start with solid data as the foundation.	needs to start with solid data as the foundation.
1:10	With the promise of churros,	With the promise of churros,
1:12	I had some friends secretly help me collect data on all 24 games	I had some friends secretly help me collect data on all 24 games
1:16	for a full day.	for a full day.
1:17	Not only did we capture how many times each game was played	Not only did we capture how many times each game was played
1:20	but also how many times people won each game	but also how many times people won each game
1:23	and the prize they won.	and the prize they won.

1:24	The first observation, based on the data we collected,	The first observation, based on the data we collected,
1:27	is that this relatively small park	is that this relatively small park
1:29	collects \$20,000 a day off their carnival games alone.	collects \$20,000 a day off their carnival games alone.
1:33	To frame the observations, we'll divide the games into three groups.	To frame the observations, we'll divide the games into three groups.
1:36	The first group, random chance games.	The first group, random chance games.
1:39	Games like these where no skill is involved	Games like these where no skill is involved
1:41	and you're basically rolling dice.	and you're basically rolling dice.
1:44	The second group, skill-based games,	The second group, skill-based games,
1:46	like the basketball shot, the milk bottle throw,	like the basketball shot, the milk bottle throw,
1:48	or the basket toss game,	or the basket toss game,
1:49	where if you bring some skill or strategy, you can increase your chances of winning.	where if you bring some skill or strategy, you can increase your chances of winning.
1:54	The final category are the games that are pretty much impossible.	The final category are the games that are pretty much impossible.
1:57	These ones are borderline scams.	These ones are borderline scams.
1:59	There are three games that lots of people attempted,	There are three games that lots of people attempted,
2:02	and nobody won.	and nobody won.
2:03	If they did win, like in the ladder climb,	If they did win, like in the ladder climb,
2:05	it was for a specific reason, which I will address later.	it was for a specific reason, which I will address later.
2:09	Let's start by talking about the random chance games.	Let's start by talking about the random chance games.
2:12	Calculating your chance of winning is straightforward.	Calculating your chance of winning is straightforward.
2:15	You divide the winning outcomes by the total outcomes	You divide the winning outcomes by the total outcomes
2:18	and get a percentage.	and get a percentage.
2:19	In this case, there are about 1,600 total cups and 160 winning cups.	In this case, there are about 1,600 total cups and 160 winning cups.
2:24	So that's a 10% chance,	So that's a 10% chance,
2:25	or one in ten throws will win.	or one in ten throws will win.
2:27	There are a ton of games similar to this.	There are a ton of games similar to this.
2:30	They use balls that are lightweight	They use balls that are lightweight
2:32	and have a high coefficient of restitution,	and have a high coefficient of restitution,
2:34	meaning they bounce well off things.	meaning they bounce well off things.
2:36	This makes it much less likely the ball will end up where you aimed it.	This makes it much less likely the ball will end up where you aimed it.

2:40	Think how much easier this game would be with beanbags,	Think how much easier this game would be with beanbags,
2:44	which are heavier and don't bounce.	which are heavier and don't bounce.
2:46	With a ping-pong ball,	With a ping-pong ball,
2:47	any imperfection in the aim of your original throw is magnified	any imperfection in the aim of your original throw is magnified
2:50	which essentially randomizes things.	which essentially randomizes things.
2:52	If you have no skills, these are the games to play.	If you have no skills, these are the games to play.
2:56	But don't get too excited, because even when you win, you lose.	But don't get too excited, because even when you win, you lose.
2:59	After some work, I uncovered where they ordered their prizes from,	After some work, I uncovered where they ordered their prizes from,
3:03	and even if you got lucky and won on your first throw,	and even if you got lucky and won on your first throw,
3:07	it would cost you \$1.50 for a prize that cost them 45 cents.	it would cost you \$1.50 for a prize that cost them 45 cents.
3:11	But you usually don't get it on your first try.	But you usually don't get it on your first try.
3:13	Treating this as an expected value problem in statistics,	Treating this as an expected value problem in statistics,
3:16	it would take you an average of five times to land it in a yellow cup,	it would take you an average of five times to land it in a yellow cup,
3:19	which means you pay \$7.50 for something that cost them 45 cents.	which means you pay \$7.50 for something that cost them 45 cents.
3:24	In the case of the big prize, by landing it in the super rare gold cup,	In the case of the big prize, by landing it in the super rare gold cup,
3:28	it's even worse.	it's even worse.
3:29	It would take you an average of 25 tries,	It would take you an average of 25 tries,
3:31	which works out to \$38 for something that cost them six bucks.	which works out to \$38 for something that cost them six bucks.
3:35	The number of people we observed winning	The number of people we observed winning
3:37	matched up pretty well with these statistical predictions.	matched up pretty well with these statistical predictions.
3:40	Let's talk about the second group. The skill-based games.	Let's talk about the second group. The skill-based games.
3:43	One of the most popular is the basketball toss,	One of the most popular is the basketball toss,
3:45	with 825 plays the day that we observed.	with 825 plays the day that we observed.
3:48	A standard three-pointer is 24 feet back on a rim that is ten feet off the ground.	A standard three-pointer is 24 feet back on a rim that is ten feet off the ground.
3:53	But in this case, the line is 28 feet back on a rim that is 11 feet off the ground,	But in this case, the line is 28 feet back on a rim that is 11 feet off the ground,
3:58	Which is subtle,	Which is subtle,

3:59	but if you have a deadly three-pointer in your muscle memory	but if you have a deadly three-pointer in your muscle memory
4:02	you will tend to miss short, which is exactly what we saw.	you will tend to miss short, which is exactly what we saw.
4:06	They have the tarp in front so that someone can't stand underneath the rim	They have the tarp in front so that someone can't stand underneath the rim
4:10	where the difference would be apparent.	where the difference would be apparent.
4:12	And again, even if you go Steph Curry and drain your first three-dollar shot,	And again, even if you go Steph Curry and drain your first three-dollar shot,
4:16	you still lose because they paid 80 cents for that basketball.	you still lose because they paid 80 cents for that basketball.
4:19	Here's another example of getting you to overestimate your chances	Here's another example of getting you to overestimate your chances
4:22	by making subtle changes, because the table is slanted up slightly	by making subtle changes, because the table is slanted up slightly
4:26	which will reduce the velocity of the ball after the bounce.	which will reduce the velocity of the ball after the bounce.
4:29	Even if you dominated this game all through college,	Even if you dominated this game all through college,
4:32	your previous experience almost becomes a handicap.	your previous experience almost becomes a handicap.
4:35	This measure-your-pitch-speed game is borderline fraudulent	This measure-your-pitch-speed game is borderline fraudulent
4:37	as their radar gun registers about 15 miles an hour too slow.	as their radar gun registers about 15 miles an hour too slow.
4:41	I know this because I measured the distance	I know this because I measured the distance
4:43	then recorded it in high speed and counted the frames.	then recorded it in high speed and counted the frames.
4:45	This pitch was clocked at 69 miles per hour,	This pitch was clocked at 69 miles per hour,
4:47	but it's much closer to 84.	but it's much closer to 84.
4:49	For the milk bottle game, the catch is the bottles are metal,	For the milk bottle game, the catch is the bottles are metal,
4:52	therefore heavier and more stable	therefore heavier and more stable
4:54	and harder to knock down than a typical bottle of that size.	and harder to knock down than a typical bottle of that size.
4:57	I've seen carnivals where these are bottom-weighted	I've seen carnivals where these are bottom-weighted
4:59	which would make them more steady and less likely to tip.	which would make them more steady and less likely to tip.
5:02	To figure out which kind you have, ask to hold it.	To figure out which kind you have, ask to hold it.
5:04	The point at which it balances on your finger is the center of mass.	The point at which it balances on your finger is the center of mass.
5:08	The key to winning this game is hitting them right here	The key to winning this game is hitting them right here

5:10	with a hard enough throw to introduce sufficient energy.	with a hard enough throw to introduce sufficient energy.
5:13	But don't throw too hard,	But don't throw too hard,
5:15	because we noticed those who threw hardest usually sacrificed on accuracy.	because we noticed those who threw hardest usually sacrificed on accuracy.
5:19	About one in 14 people knock this over on their first try.	About one in 14 people knock this over on their first try.
5:22	Finally, the basket toss game.	Finally, the basket toss game.
5:23	The key is to have your first bounce hit on this front lip	The key is to have your first bounce hit on this front lip
5:27	to reduce the kinetic energy enough that it won't bounce back out.	to reduce the kinetic energy enough that it won't bounce back out.
5:30	About one in ten throws will win on this game, according to our observations.	About one in ten throws will win on this game, according to our observations.
5:34	This brings us to the final category of the near-impossible games.	This brings us to the final category of the near-impossible games.
5:37	There are three of them.	There are three of them.
5:38	On this one, the goal is to shoot out this red star completely	On this one, the goal is to shoot out this red star completely
5:42	with this automatic BB gun.	with this automatic BB gun.
5:43	The best strategy is to shoot a circle around the star to cut it out.	The best strategy is to shoot a circle around the star to cut it out.
5:47	Not only are the guns not accurate or precise,	Not only are the guns not accurate or precise,
5:50	the bigger issue is you start out doing well	the bigger issue is you start out doing well
5:53	because there's enough surrounding paper for the BB to easily rip through.	because there's enough surrounding paper for the BB to easily rip through.
5:57	But Newton's third law tells us that you can only push on something	But Newton's third law tells us that you can only push on something
6:00	as hard as it can resist your push.	as hard as it can resist your push.
6:02	At the end, you have these barely supported pieces of the star	At the end, you have these barely supported pieces of the star
6:05	that move out of the way when the BB comes	that move out of the way when the BB comes
6:07	without building up enough stress to rip the paper.	without building up enough stress to rip the paper.
6:09	Out of 120 plays, we saw nobody win this game all day.	Out of 120 plays, we saw nobody win this game all day.
6:14	This ring bottle game is also impossibly difficult.	This ring bottle game is also impossibly difficult.
6:17	It's a lightweight object that's bouncy to encourage randomness,	It's a lightweight object that's bouncy to encourage randomness,
6:21	but the actual inner diameter of the ring	but the actual inner diameter of the ring
6:23	is really close to the outer diameter of the bottle.	is really close to the outer diameter of the bottle.

6:26	This means that any throw, besides this pretty much perfect one,	This means that any throw, besides this pretty much perfect one,
6:29	will send the ring bouncing away without settling on the bottle.	will send the ring bouncing away without settling on the bottle.
6:32	If you really want this bear, I suggest going on Amazon and getting it for \$47.	If you really want this bear, I suggest going on Amazon and getting it for \$47.
6:37	I can't tell you how much money this will save you,	I can't tell you how much money this will save you,
6:40	'cause of the 840 rings we saw thrown, none stayed on a bottle.	'cause of the 840 rings we saw thrown, none stayed on a bottle.
6:43	Which brings up an obvious rule of thumb.	Which brings up an obvious rule of thumb.
6:45	If you want the feeling of winning a game,	If you want the feeling of winning a game,
6:47	do not stop at any booth that offers really big prizes.	do not stop at any booth that offers really big prizes.
6:51	For the final near-impossible game,	For the final near-impossible game,
6:53	let me reiterate that the most lucrative games	let me reiterate that the most lucrative games
6:56	are those in which the customers overestimate their chances of success.	are those in which the customers overestimate their chances of success.
6:59	No game is a better example of that than the ladder climb.	No game is a better example of that than the ladder climb.
7:02	There's a subtle issue that I think people realize	There's a subtle issue that I think people realize
7:05	but don't internalize the significance.	but don't internalize the significance.
7:07	The ladder converges to be supported on the wall at one point instead of two.	The ladder converges to be supported on the wall at one point instead of two.
7:11	If it was attached at two points it's like crawling across a rope bridge,	If it was attached at two points it's like crawling across a rope bridge,
7:15	which is pretty easy.	which is pretty easy.
7:17	Let's pretend this is you.	Let's pretend this is you.
7:18	If you shrunk the weight of your body down to the average location,	If you shrunk the weight of your body down to the average location,
7:21	we call that spot the center of mass, which we'll mark with this dot.	we call that spot the center of mass, which we'll mark with this dot.
7:25	And we can double-check this is the right spot	And we can double-check this is the right spot
7:28	because it balances perfectly on one finger.	because it balances perfectly on one finger.
7:30	If you draw imaginary lines connecting the different support points,	If you draw imaginary lines connecting the different support points,
7:34	that creates an area.	that creates an area.
7:35	As long as your center-of-mass dot is within that area,	As long as your center-of-mass dot is within that area,
7:38	it's impossible to fall off.	it's impossible to fall off.
7:40	As soon as your center-of-mass dot is even a little bit outside the area,	As soon as your center-of-mass dot is even a little bit outside the area,

7:44	you start to rotate and fall off.	you start to rotate and fall off.
7:46	This is true no matter which way you orient it.	This is true no matter which way you orient it.
7:48	If you've ever bent over to pick something up, you know this fact	If you've ever bent over to pick something up, you know this fact
7:51	whether you realize or not.	whether you realize or not.
7:52	The region of support is between the back of your heels	The region of support is between the back of your heels
7:55	and the tips of your toes.	and the tips of your toes.
7:56	When you reach over, you will naturally move your butt back	When you reach over, you will naturally move your butt back
7:59	to keep your center of mass in between those support points.	to keep your center of mass in between those support points.
8:02	Try picking something up while standing against a wall	Try picking something up while standing against a wall
8:05	so you can't move your butt back.	so you can't move your butt back.
8:07	The moment your center of mass gets beyond your toes,	The moment your center of mass gets beyond your toes,
8:10	you start to tip over.	you start to tip over.
8:11	In the ladder game, you're only connected at one point.	In the ladder game, you're only connected at one point.
8:14	Even though it looks wide because of the rungs,	Even though it looks wide because of the rungs,
8:17	that area of support reduces down to a line.	that area of support reduces down to a line.
8:20	If you don't keep your center of mass above that line,	If you don't keep your center of mass above that line,
8:22	you will start to rotate and fall off.	you will start to rotate and fall off.
8:24	In other words, to win this game,	In other words, to win this game,
8:26	you need to be able to crawl across a tightrope.	you need to be able to crawl across a tightrope.
8:29	You might think, "I can do a slackline, so I can do this,"	You might think, "I can do a slackline, so I can do this,"
8:32	but a slackline is much easier for two reasons.	but a slackline is much easier for two reasons.
8:35	You can flail your arms and legs to adjust your center of mass	You can flail your arms and legs to adjust your center of mass
8:38	to keep it directly above that line of support.	to keep it directly above that line of support.
8:40	Your center of mass is higher, increasing your mass moment of inertia	Your center of mass is higher, increasing your mass moment of inertia
8:43	making you more stable,	making you more stable,
8:45	in the same way it's easier to balance this umbrella when it's extended	in the same way it's easier to balance this umbrella when it's extended
8:48	versus when it's collapsed.	versus when it's collapsed.

8:50	There are videos that say tricks like maintain three points of contact.	There are videos that say tricks like maintain three points of contact.
8:54	They're all useless,	They're all useless,
8:55	because keeping your center of mass directly above a line	because keeping your center of mass directly above a line
8:58	is something you have to get a feel for that takes a lot of practice.	is something you have to get a feel for that takes a lot of practice.
9:02	Once you've had enough practice,	Once you've had enough practice,
9:03	this is the one game at the carnival that's all skill	this is the one game at the carnival that's all skill
9:06	so you can win every time and clean them out.	so you can win every time and clean them out.
9:09	Unfortunately, the owners know this	Unfortunately, the owners know this
9:11	which is why it's also the only game with this super lame caveat.	which is why it's also the only game with this super lame caveat.
9:14	Now that we were experts, I called up Matt Winaker,	Now that we were experts, I called up Matt Winaker,
9:17	who was recently drafted to play baseball for the New York Mets,	who was recently drafted to play baseball for the New York Mets,
9:20	to maximize our chances of winning any skill game to do with throwing.	to maximize our chances of winning any skill game to do with throwing.
9:44	Clearly, Matt had a deadly lock on any throwing game,	Clearly, Matt had a deadly lock on any throwing game,
9:48	but basketball's more my game.	but basketball's more my game.
9:49	So to finish off the day, I decided I'd bring his ego into check.	So to finish off the day, I decided I'd bring his ego into check.
9:53	But as it turns out, if you are a world-class athlete in one sport,	But as it turns out, if you are a world-class athlete in one sport,
9:56	you are a really, really good athlete in all sports.	you are a really, really good athlete in all sports.
9:59	In conclusion, play the games if you think they're fun.	In conclusion, play the games if you think they're fun.
10:02	Just know the odds are stacked against you, so if you lose, it's NBD.	Just know the odds are stacked against you, so if you lose, it's NBD.
10:06	Unlike this guy, who lost his life savings playing carnival games.	Unlike this guy, who lost his life savings playing carnival games.
10:10	If your motivation is to gain the love and admiration of someone special	If your motivation is to gain the love and admiration of someone special
10:13	via a stuffed animal, you don't need carnival games to do that.	via a stuffed animal, you don't need carnival games to do that.
10:17	Amazon works just as well.	Amazon works just as well.
10:19	I just bought this for you, my lady.	I just bought this for you, my lady.
10:24	Why?	Why?